

Barcode cards, check-digit rule, and answer key

The UPC-A check-digit rule

A product barcode has 12 digits. The last one is a **check digit** computed from the first 11 by a fixed rule, so a single mistyped digit no longer matches and the scanner rejects the code.

- 01 Add the digits in the odd positions (1st, 3rd, 5th, ...) and multiply that sum by 3.
- 02 Add the digits in the even positions (2nd, 4th, ...).
- 03 Add those two results, then find what you must add to reach the next multiple of 10. That is the check digit.

Worked example for **0 36000 29145**: the rule gives a check digit of **2**, so the full valid code is 036000291452. Card 1 uses it.

1 0 36000 29145 2 ,

2 0 12445 67890 5 ,

3 0 40123 45678 0 ,

4 0 73500 54850 2 ,

5 9 84912 01203 6 ,

6 0 12000 80351 2 ,

7 0 50002 59407 7 ,

8 0 38000 12345 0 ,

9 0 97804 71496 8 ,

10 0 64144 10062 0

11 0 17500 00201 7

12 0 07541 92014 2

Instructor answer key (do not hand out)

Keep this page with the instructor. Each corrupted card has exactly one wrong digit, which the mod-10 rule always catches. Reject the corrupted codes; accept the rest.

CARD	CODE	VERDICT
1	0 36000 29145 2	Valid — accept
2	0 12445 67890 5	Corrupted — reject
3	0 40123 45678 0	Valid — accept
4	0 73500 54850 2	Corrupted — reject
5	9 84912 01203 6	Corrupted — reject
6	0 12000 80351 2	Valid — accept
7	0 50002 59407 7	Corrupted — reject
8	0 38000 12345 0	Valid — accept
9	0 97804 71496 8	Corrupted — reject
10	0 64144 10062 0	Valid — accept
11	0 17500 00201 7	Corrupted — reject
12	0 07541 92014 2	Corrupted — reject